Henintsoa Harinoro Randrianasolo

+261 34 94 951 65 | henintsoa1001@gmail.com | linkedin.com/in/henintsoa | github.com/HenintsoaHARINORO

Professional Summary

AI Engineer with a background in artificial intelligence and electronics engineering, specializing in data-driven systems and predictive modeling. Experienced in designing end-to-end data pipelines, developing machine learning models, and building scalable software systems. Strong interest in industrial systems, energy applications, and decision-support tools for predictive maintenance.

Technical Skills

AI Frameworks & LLMs: LangChain, RAG, Prompt Engineering (CoT), Fine-tuning (LoRA), Hugging Face, Ollama
Development & APIs: Python, FastAPI, Java, Streamlit, RESTful APIs, Vector Embeddings
Data & Vector Systems: PostgreSQL, MySQL, Pandas, NumPy, Knowledge Retrieval Systems
DevOps & Deployment: Docker, Git, CI/CD, Linux, Cloud Integration, n8n Automation

Professional Experience

AI Developer Apr 2026 – Present
Bay Consulting *Antananarivo, MG*
- Designed and architected autonomous AI agents to automate complex, multi-step workflows.
- Developed a hybrid RAG pipeline for the ingestion and structured extraction of financial product catalogs.

Intern AI Engineer Dec 2024 – Dec 2025
Giris *Remote*
- Designed and deployed AI-driven systems for automated decision-making, including data pipelines for real-time information processing and retrieval.
- Built and deployed scalable AI microservices using FastAPI and Docker, ensuring reliable production environments.
- Engineered domain-specific LLMs by fine-tuning models with LoRA, significantly improving accuracy for specialized use cases.
- Developed an intelligent automation system using FastText to automate data quality checks and spelling corrections.

Data Analyst Jul 2023 – Mar 2025
Yas Madagascar *Antananarivo, MG*
- Developed predictive models to forecast trends using historical data, including data cleaning, feature engineering, and model validation.
- Created data-driven product insights by tracking user engagement via Google Analytics.

Personal Projects

Bible Daily Automation | *n8n, Ollama, PostgreSQL, Docker*
- Developed an automated pipeline that fetches scripture, generates AI explanations, converts them to speech, and delivers both text and audio to Telegram. *[Link]*

Web Scraping and RAG Application | *Python, LangChain, RAG*
- Built an AI system leveraging web scraping to collect company data and generate professional summaries with LLMs, supporting natural-language querying. *[Link]*

Chat with PDF (RAG Application) | *Streamlit, Ollama, RAG*
- Built an interactive application enabling users to query PDF documents with context-aware answers via a RAG pipeline.

Education

Moscow Institute of Physics and Technology (MIPT) Oct 2022 – Jun 2024
Master of Science, Modern State of Artificial Intelligence *Grade: 4.50/7.00*
- **Thesis:** Math problem solving using the Chain of Thought (CoT) *(Advisor: Radoslav Neychev)*
- Evaluated mathematical reasoning of LLMs by implementing and testing CoT prompting methods. *[Link]*

Ecole Supérieure Polytechnique d'Antananarivo Jun 2017 – Dec 2023
Engineering Degree, Electronics Engineering *Antananarivo, MG*
- **Thesis:** "Classification de Mémoire Numérique par la technologie NLP" *(Advisor: Alain Tovohery)*
- Developed a multi-class classification model using Doc2Vec and Logistic Regression to categorize documents (79% precision).

Certifications & Awards

First Prize, Hack The Box 2024 (Axian Group): Created RESAO, a localized voice recognition app for navigation.
SMILES (Summer of Machine Learning) Skoltech: Research on Transformers vs RNNs (GRU/GPT-based models). *[Link]*
Google Analytics for Beginners; Data Science Math Skills (Coursera)